

Hypotenuse berechnen

Klasse 9 · Lösungen

1a) c

b) r

c) s

d) e

2a) $w^2 = u^2 + v^2$

Hypotenuse w

b) $x^2 = y^2 + z^2$

Hypotenuse x

c) $\overline{BC}^2 = \overline{AB}^2 + \overline{AC}^2$

rechter Winkel bei A , Hypotenuse $\overline{BC}$

d) gilt nicht

kein rechter Winkel

3b) $c = 13 \text{ cm}$

$$c^2 = 5^2 + 12^2 = 169 \Rightarrow c = 13; 12 < 13 < 17$$

c) $c = 15 \text{ cm}$

$$c^2 = 9^2 + 12^2 = 225 \Rightarrow c = 15$$

4a) $c = \sqrt{97} \approx 9,8 \text{ cm}$

$$c^2 = 4^2 + 9^2 = 97 \Rightarrow c \approx 9,849$$

b) $c = 130 \text{ cm}$

$$1,2 \text{ m} = 120 \text{ cm}; c^2 = 120^2 + 50^2 = 16\,900$$

5. $d = 51 \text{ cm}$

$$d^2 = 45^2 + 24^2 = 2\,601$$

6. 40 m

$$\text{quer: } \sqrt{80^2 + 60^2} = 100 \text{ m}; 140 - 100 = 40$$

7. nein

$$\text{Diagonale: } \sqrt{40^2 + 30^2} = 50 \text{ cm} < 52 \text{ cm}$$

8. $\overline{AB} = \sqrt{1\,300} \approx 36,1 \text{ m}$

$$\overline{AB}^2 = 12^2 + 34^2 = 1\,300 \Rightarrow \overline{AB} \approx 36,06$$

9. $x = \sqrt{29\,156} \approx 170,8 \text{ cm}$

$$x^2 = 170^2 + 16^2 = 29\,156 \Rightarrow x \approx 170,75$$